

Day – 6 Switching Basics Theory.

Task – 1 Create topology with two pc and one switch.

- Step 1: Build the Topology :- Select Switches from the bottom-left menu and place a 2960 Switch (or similar model) in the workspace.
- Select End Devices and place two PCs (PC0 and PC1) next to the switch.
- Connect the Devices:
- Click the Connections icon (lightning bolt).
- Choose the Copper Straight-Through cable.
- Connect PC0 to the switch's FastEthernet0/1 port.
- Connect PC1 to the switch's FastEthernet0/2 port.
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- Step 2: Access the Switch CLI
- Click on the Switch to open its configuration window.
- Navigate to the CLI (Command Line Interface) tab.
- Press Enter to get the command prompt, then enter privileged EXEC mode:
- Enable
-
- Step 3: Check the MAC Address Table
- Run the command to view the switch's forwarding database

Task – 2 Sending ping from pc1 to pc2 to simulate network.

- Step 1: Assign IP Addresses to the PCs:- Before pinging, ensure both PCs are on the same subnet:
- PC1: IP Address 192.168.1.10, Subnet Mask 255.255.255.0
- PC2: IP Address 192.168.1.20, Subnet Mask 255.255.255.0
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- Step 2: Send the Ping:- Click on PC1, go to the Desktop tab, and open the Command Prompt.
- Run the ping command to reach PC2:
- ping 192.168.1.20
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- Step 3: Check the MAC Address Table Again:- Immediately after sending the ping, go back to the Switch CLI and run the command:
- show mac address-table

- Now, entries populated in the table, similar to this:
- Mac Address Table
- -----
- Vlan Mac Address Type Ports
- -----
- 1 0001.64ab.cdef DYNAMIC Fa0/1
- 1 00e0.f712.3456 DYNAMIC Fa0/2
- -----
- Total Mac Addresses for this criterion: 2

Task – 3 Add and check the pc3 to pc1.

- Step 1: Add and Connect PC3
- Place the Device: Drag and drop a third PC (PC3) from the End Devices menu into your Packet Tracer workspace.
- Make the Connection: Select the Copper Straight-Through cable from the Connections menu and connect PC3 to the switch's next available port, FastEthernet0/3.
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- Step 2: Assign an IP Address to PC3
- Click on PC3, navigate to the Desktop tab, open IP Configuration, and set its IP details on the same subnet as the other PCs:
- IP Address: 192.168.1.30
- Subnet Mask: 255.255.255.0
- Default Gateway:
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- Step 3: Send a Ping from PC3 to PC1
- Click on PC3, go to its Desktop tab, and open the Command Prompt.
- Ping PC1 (192.168.1.10):
- ping 192.168.1.10
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- Step 4: Observe the Updated MAC Address Table
- Immediately after the ping completes, go to the Switch CLI and run the command:
- show mac address-table
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- The MAC address table will now contain three entries, showing that the switch has successfully learned PC3's MAC address on port Fa0/3 while keeping the existing entries for PC1 and PC2:
- Plaintext
- Mac Address Table

- -----
- Vlan Mac Address Type Ports
- -----
- 1 0001.64ab.cdef DYNAMIC Fa0/1 <-- PC1
- 1 00e0.f712.3456 DYNAMIC Fa0/2 <-- PC2
- 1 0050.56aa.bbcc DYNAMIC Fa0/3 <-- PC3 (New entry)
- -----
- Total Mac Addresses for this criterion: 3

Task – 4 Explain functioning in the port 1 and port 2.

- Step 1: Frame Arrival and Source Learning:- The switch receives the frame on Port 1.
- As standard practice, the switch inspects the source MAC address of the incoming frame to ensure its MAC address table is up to date, confirming that 00:11:22:33:44:55 is located on Port 1.
- Step 2: Destination MAC Lookup:- The switch next looks at the destination MAC address in the frame's header, which is 66:77:88:99:AA:BB.
- It searches its MAC address table (CAM table) to find a matching entry for this destination address.
- Step 3: Table Match:- The switch finds a match in its table: MAC address 66:77:88:99:AA:BB is explicitly mapped to Port 2.
- Step 4: Forwarding Decision:- Because the destination port (Port 2) is different from the port where the frame entered (Port 1), the switch forwards the frame out of Port 2.
- It does *not* flood the frame to other ports, ensuring that communication remains targeted and efficient between the two endpoints.

Task – 5 Diagrams which show switch learns on Mac address.

